The documentation process conversion measures necessary to comply with this revision shall be completed by 28 October 1999

INCH-POUND

MIL-PRF-19500/560C
28 July 1999
SUPERSEDING
MIL-S-19500/560B
2 September 1994

PERFORMANCE SPECIFICATION SHEET
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING
TYPE 2N5339 JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon switching transistors. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500 and two levels of product assurance are provided for each unencapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-39), and figures 2 and 3 for JANHC and JANKC (die) dimensions.

1.3 Maximum ratings.

| $P_T$ 1/ $T_A$ = +25°C | $P_T$ 2/ $T_C$ = +25°C | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_C$ | $I_B$ | $T_{OP}$ and $T_{STG}$ | $R_{\theta JC}$ |
|---|---|---|---|---|---|---|---|---|
| W | W | V dc | V dc | V dc | A dc | A dc | °C | °C/W |
| 1.0 | 10 | 100 | 100 | 6.0 | 5.0 | 1.0 | -55 to +200 | 17.5 |

1/ Derate linearly at 5.71 mW/°C above $T_A$ > +25°C.
2/ Derate linearly at 57.1 mW/°C above $T_C$ > +25°C.

1.4 Primary electrical characteristics $T_A$ = +25°C. (Unless otherwise indicated, applies to all devices.)

| Limits | $h_{FE1}$ 1/ $V_{CE}$ = 2.0 V dc; $I_C$ = 0.5 A dc | $h_{FE2}$ 1/ $V_{CE}$ = 2.0 V dc; $I_C$ = 2.0 A dc | $h_{FE3}$ 1/ $V_{CE}$ = 2.0 V dc; $I_C$ = 5.0 A dc |
|---|---|---|---|
| Min | 60 | 60 | 40 |
| Max | | 240 | |

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FSC 5961

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1.4 Primary electrical characteristics $T_A$ = +25°C. (Unless otherwise indicated, applies to all devices.) - Continued.

| Limits | $\|h_{FE}\|$<br>f = 10 MHz<br>$V_{CE}$ = 10 V dc<br>$I_C$ = 0.5 A dc | $C_{obo}$<br>$V_{CE}$ = 10 V dc<br>$I_E$ = 0<br>100 kHz ≤ f ≤ 1 MHz | Switching | | $V_{CE(SAT)1}$<br>$I_C$ = 2.0 A dc<br>$I_B$ = 0.2 A dc<br>1/ | $V_{BE(SAT)1}$<br>$I_C$ = 2.0 A dc<br>$I_B$ = 0.2 A dc<br>1/ |
|---|---|---|---|---|---|---|
| | | | See figure 4<br>$t_{on}$ | See figure 5<br>$t_{off}$ | | |
| | | pF | µs | µs | V dc | V dc |
| Min | 3.0 | | | | | |
| Max | 15.0 | 250 | 0.2 | 2.2 | 0.7 | 1.2 |

1/ Pulsed (see 4.5.1).

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.



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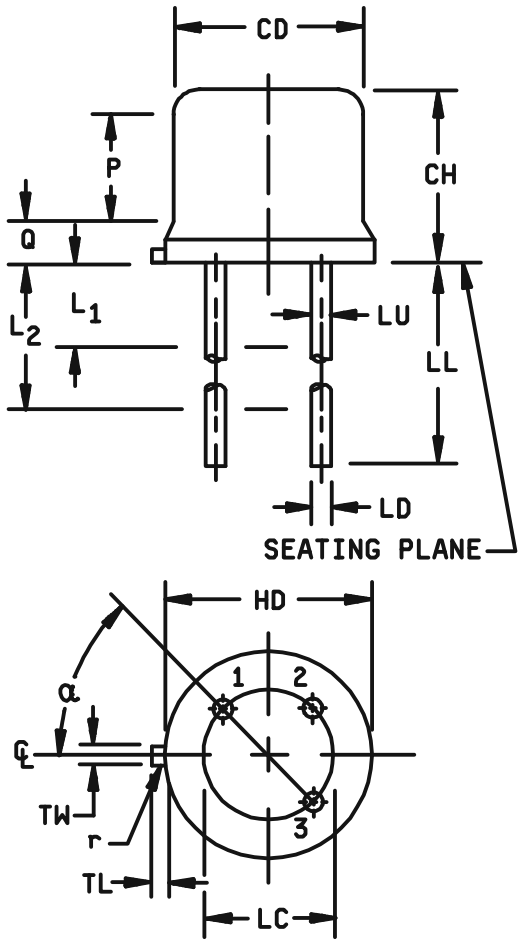



FIGURE 1. Physical dimensions (TO-39).



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | 0.305 | 0.355 | 7.75 | 8.51 | |
| CH | 0.240 | 0.260 | 6.10 | 6.60 | |
| HD | 0.335 | 0.370 | 8.51 | 9.40 | |
| LC | 0.200 TP | | 5.08 TP | | 6 |
| LD | 0.016 | 0.021 | 0.41 | 0.53 | 7 |
| LL | 0.500 | 0.750 | 12.70 | 19.05 | 7 |
| LU | 0.016 | 0.019 | 0.41 | 0.48 | 7 |
| L1 | | 0.050 | | 1.27 | 7 |
| L2 | 0.250 | | 6.35 | | 7 |
| TL | 0.029 | 0.045 | 0.74 | 1.14 | 3 |
| TW | 0.028 | 0.034 | 0.71 | 0.86 | 10 |
| P | 0.100 | | 2.54 | | 5 |
| Q | | 0.050 | | 1.27 | 4 |
| r | | 0.010 | | 0.25 | 11 |
| α | 45° TP | | 45° TP | | 6 |
| Notes | 1, 2, 8, 9 | | 1, 2, 8, 9 | | |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Details of outline in this zone are optional.
5. Symbol CD shall not vary more than 0.010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane 0.054 inch (1.37 mm) +0.001 inch (0.03 mm) -0.000 inch (0.00 mm) below seating plane shall be within 0.007 inch (0.18 mm) radius of true position (TP) relative to tab. Device may be measured by direct methods or by gauge.
7. Symbol LD applies between $L_1$ and $L_2$. Dimension LD applies between $L_2$ and LL minimum.
8. Lead designation, depending on device type, shall be as follows:

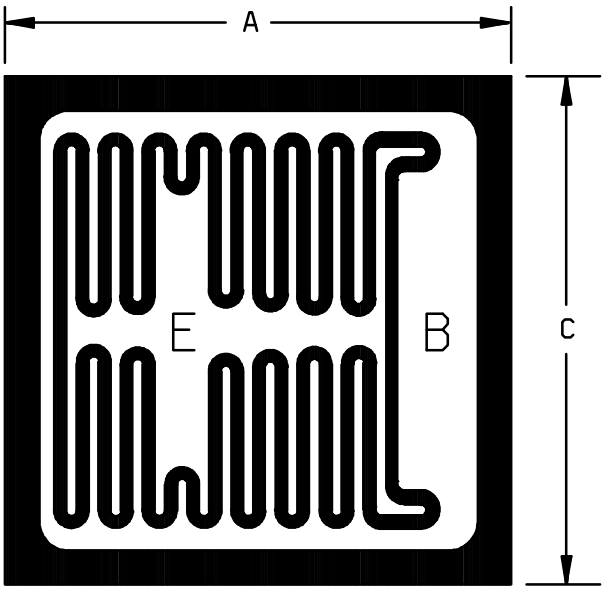
| Lead number | TO-39 |
|---|---|
| 1 | Emitter |
| 2 | Base |
| 3 | Collector |

9. Lead number three is electrically connected to case.
10. Beyond r maximum, TW shall be held for a minimum length of 0.011 inch (0.28 mm).
11. Symbol r applied to both inside corners of tab.

FIGURE 1. Physical dimensions (TO-39) continued.



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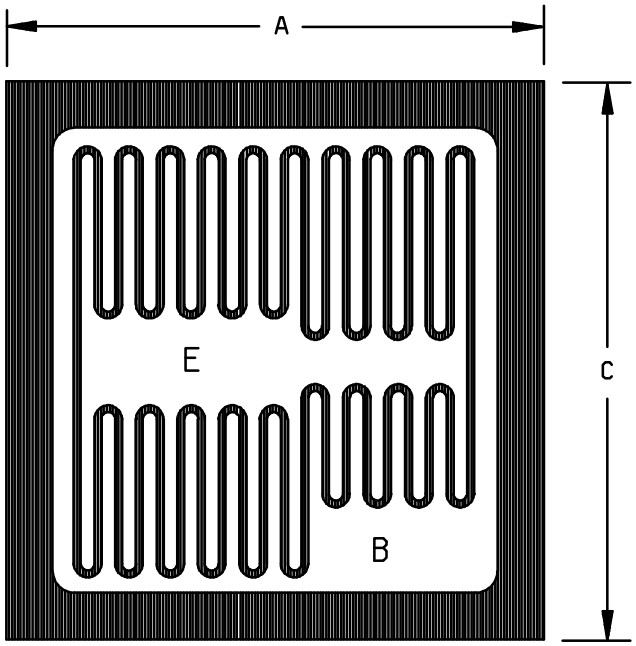

| Letter | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | 0.120 | 0.124 | 3.05 | 3.15 |
| C | 0.120 | 0.124 | 3.05 | 3.15 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Thickness: 0.008 inch (0.20 mm) to 0.012 inch (0.30 mm).
   Top metal: Aluminum 40,000 Å minimum, 50,000 Å nominal.
   Back metal: Gold 2,500 Å minimum, 3,000 Å nominal.
   Back side: Collector.
   Bonding pad: B = 0.015 inch (0.38 mm) x 0.072 inch (1.83 mm).
   E = 0.015 inch (0.38 mm) x 0.060 inch (1.52 mm).
4. Unless otherwise specified, tolerance is ± 0.005 inch (0.13 mm).

FIGURE 2. Physical dimensions JANHCA and JANKCA.



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| Letter | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | 0.098 | 0.102 | 2.49 | 2.59 |
| C | 0.098 | 0.102 | 2.49 | 2.59 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Thickness: 0.006 inch (0.15 mm) to 0.010 inch (0.25 mm).
   Top metal: Aluminum 25,000 Å minimum, 33,000 Å nominal.
   Back metal: Gold 1,500 Å minimum, 2,000 Å nominal.
   Back side: Collector.
4. Unless otherwise specified, tolerance is ± 0.005 inch (0.13 mm).

FIGURE 3. Physical dimensions JANHCB and JANKCB.



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3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (TO-39) and figures 2 and 3 (JANHC and JANKC) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Devices shall be marked in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and 6.3 herein.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see MIL-PRF-19500) | Measurements | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| 9 | $I_{CBO1}$ and $h_{FE2}$ | Not applicable |
| 11 | $I_{CBO1}$; $h_{FE2}$,<br>$\Delta I_{CBO1} = \pm$ 100 percent of initial value or 1.0 µA dc, whichever is greater;<br>$\Delta h_{FE2} = \pm$ 15 percent | $I_{CBO1}$ and $h_{FE2}$ |
| 12 | See 4.3.1 | See 4.3.1 |
| 13 | Subgroups 2 and 3 of table I herein;<br>$\Delta I_{CBO1} = \pm$ 100 percent of initial value or 1.0 µA dc, whichever is greater;<br>$\Delta h_{FE2} = \pm$ 15 percent | Subgroup 2 of table I herein;<br>$\Delta I_{CBO1} = \pm$ 100 percent of initial value or 1.0 µA dc, whichever is greater;<br>$\Delta h_{FE2} = \pm$ 15 percent. |

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

$V_{CB} \geq$ 10 V dc, $P_T$ = 1.0 W at $T_A$ = Room ambient as defined in the general requirements of MIL-STD-750.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Screening for JANHC and JANKC. Screening for JANHC and JANKC see MIL-PRF-19500.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.



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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein. . Electrical measurements (end-points) requirements shall be in accordance with group A, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa of MIL-PRF-19500 (JANS) and 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with group A, subgroup 2 herein. See 4.4.2.2 herein and table VIb of MIL-PRF-19500 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) requirements shall be in accordance with group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B4 | 1037 | $V_{CB} \geq 10$ V dc, $P_T = 1.0$ W; 2,000 cycles. |
| B5 | 1027 | $V_{CB} \geq 10$ V dc, $T_A = +125^{\circ}C \pm 25^{\circ}C$ for 96 hours or $T_A = +100^{\circ}C$ for 96 hours with $P_T$ adjusted as required by the chosen $T_A$ to give an average lot. $T_J = +275^{\circ}C$. No heat sink shall be permitted. |
| B5 | 2037 | Test condition A. |

4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1037 | For solder die attach: $V_{CB} \geq 10$ V dc, $T_A$ = Room ambient as defined in the general requirements of MIL-STD-750. 2,000 cycles. |
| B3 | 1027 | For eutectic die attach: $V_{CB} \geq 10$ V dc, Adjust $P_T$ to achieve $T_J = +150^{\circ}C$ minimum. $T_A$ = Room ambient as defined in the general requirements of MIL-STD-750. |
| B6 | 1032 | $T_A = +200^{\circ}C$. |
| B7 | 3071 | High-temperature operation; $T_A = +150^{\circ}C$. Collector to emitter voltage (saturated); $I_C$ 2.0 A dc, $I_B$ = 0.2 A dc, (pulsed, see 4.5.1); $\Delta V_{CE(sat)3}$, maximum limit 0.75 V dc. |
| | 3066 | Low-temperature operation; $T_A = -55^{\circ}C$. Base to emitter voltage (saturated); test condition A, $I_C$ 2.0 A dc, $I_B$ = 0.2 A dc, (pulsed, see 4.5.1); $\Delta V_{BE(sat)3}$, maximum limit 1.3 V dc. Sampling plan: 22 devices, c = 0. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and herein. Electrical measurements (end points) shall be in accordance with group A, subgroup 2 herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E. |
| C6 | 1037 | For solder die attach: $V_{CB} \geq 10$ V dc, $T_A$ = Room ambient as defined in the general requirements of MIL-STD-750. 6,000 cycles. |
| C6 | 1027 | For eutectic die attach: $V_{CB} \geq 10$ V dc, Adjust $P_T$ to achieve $T_J = +150^{\circ}C$ minimum. $T_A$ = Room ambient as defined in the general requirements of MIL-STD-750. |

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500, and herein. Electrical measurements (end points) shall be in accordance with group A, subgroup 2 herein.



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4.4.3.1 Group E inspection, table IX of MIL-PRF-19500.

| Subgroup | Method | Condition | Sampling plan |
|---|---|---|---|
| E1 | 1051 | 500 cycles<br>Electrical measurements<br>See table I, subgroup 2. | 22 devices, c = 0 |
| E2 | 1039 | Condition A, 340 hours<br>Electrical measurements<br>See table I, subgroup 2. | 22 devices, c = 0 |
| E3 | | Not applicable | |
| E4 | 3131 | See 4.5.3 | 22 devices, c = 0 |

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.3 Thermal resistance (to be performed for qualification inspection only). The thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 0.15 A dc.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be + 25°C ≤ $T_R$ ≤ +35° C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit shall be $R_{\theta JC}$ = 17.5° C/W.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D; $I_C$ = 50 mA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 100 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition D; $V_{CE}$ = 100 V dc | $I_{CEO}$ | | 100 | µA dc |
| Collector to emitter cutoff current | 3041 | Bias condition A; $V_{BE}$ = 1.5 V dc; $V_{CE}$ = 90 V dc | $I_{CEX1}$ | | 10 | µA dc |
| Collector to base cutoff current | 3036 | Bias condition D; $V_{CB}$ = 100 V dc | $I_{CBO}$ | | 10 | µA dc |
| Emitter to base, cutoff current | 3061 | Bias condition D; $V_{EB}$ = 6.0 V dc | $I_{EBO}$ | | 100 | µA dc |
| Forward - current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 0.5 A dc, pulsed (see 4.5.1) | $h_{FE1}$ | 60 | | |
| Forward - current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 2.0 A dc; pulsed (see 4.5.1) | $h_{FE2}$ | 60 | 240 | |
| Forward - current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 5.0 A dc; pulsed (see 4.5.1) | $h_{FE3}$ | 40 | | |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 2.0 A dc; $I_B$ = 0.2 A dc; pulsed (see 4.5.1) | $V_{CE(SAT)1}$ | | 0.7 | V dc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 5.0 A dc; $I_B$ = 0.5 A dc; pulsed (see 4.5.1) | $V_{CE(SAT)2}$ | | 1.2 | V dc |
| Base to emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 2.0 A dc; $I_B$ = 0.2 A dc; pulsed (see 4.5.1) | $V_{BE(SAT)1}$ | | 1.2 | V dc |
| Base to emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 5.0 A dc; $I_B$ = 0.5 A dc; pulsed (see 4.5.1) | $V_{BE(SAT)2}$ | | 1.8 | V dc |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High - temperature operation | | $T_A$ = +150°C | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition A; $V_{CE}$ = 90 V dc; $V_{BE}$ = 1.5 V dc; | $I_{CEX2}$ | | 1.0 | mA dc |
| Low - temperature operation | | $T_A$ = -55°C | | | | |
| Forward - current transfer ratio | 3076 | $V_{CE}$ = 2.0 V dc; $I_C$ = 2.0 A dc; Pulsed (see 4.5.1) | $h_{FE4}$ | 12 | | |
| Subgroup 4 | | | | | | |
| Small - signal short - circuit forward - current transfer ratio | 3306 | $V_{CE}$ = 10 V dc; $I_C$ = 0.5 A dc; f = 10 MHz | $\|h_{fe}\|$ | 3 | 15 | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 250 | pF |
| Input capacitance (output open - circuited) | 3240 | $V_{BE}$ = 2.0 V dc; $I_C$ = 0; 100 kHz ≤ f ≤ 1 MHz (see 4.5.2) | $C_{ibo}$ | | 1,000 | pF |
| Pulse response | | | | | | |
| Pulse delay time | 3251 | See figure 4 | $t_d$ | | 100 | ns |
| Pulse rise time | 3251 | See figure 4 | $t_r$ | | 100 | ns |
| Pulse storage time | 3251 | See figure 5 | $t_s$ | | 2 | µs |
| Pulse fall time | 3251 | See figure 5 | $t_f$ | | 200 | ns |

See footnote at end of table.



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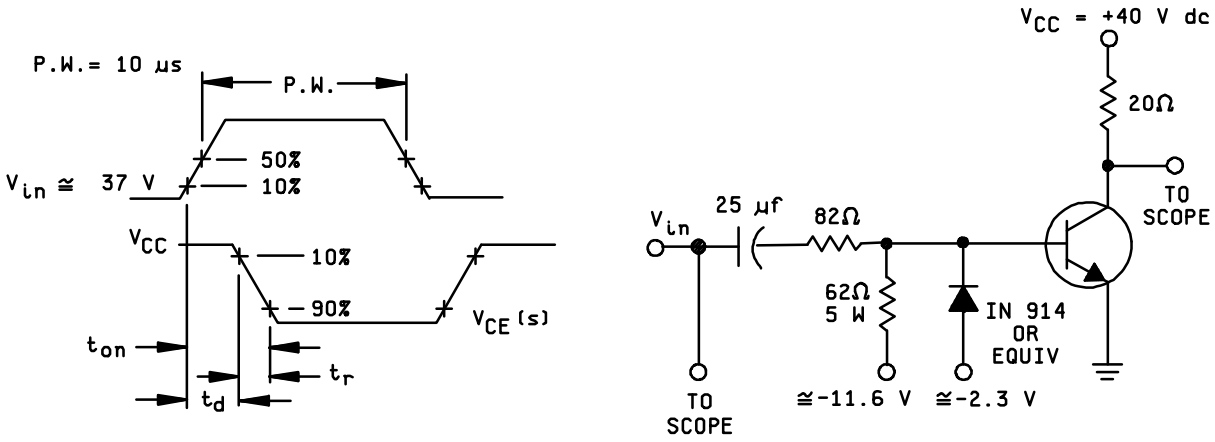
TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 5 | | | | | | |
| Safe operating area (continuous dc) | 3051 | $T_C$ = +25°C; t ≥ 0.5 s; 1 cycle | | | | |
| Test 1 | | $V_{CE}$ = 2.0 V dc; $I_C$ = 5.0 A dc | | | | |
| Test 2 | | $V_{CE}$ = 5.0 V dc; $I_C$ = 2.0 A dc | | | | |
| Test 3 | | $V_{CE}$ = 90 V dc; $I_C$ = 55 mA dc | | | | |
| End-point electrical measurements | | See table I, group A, subgroup 2 | | | | |
| Subgroups 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500



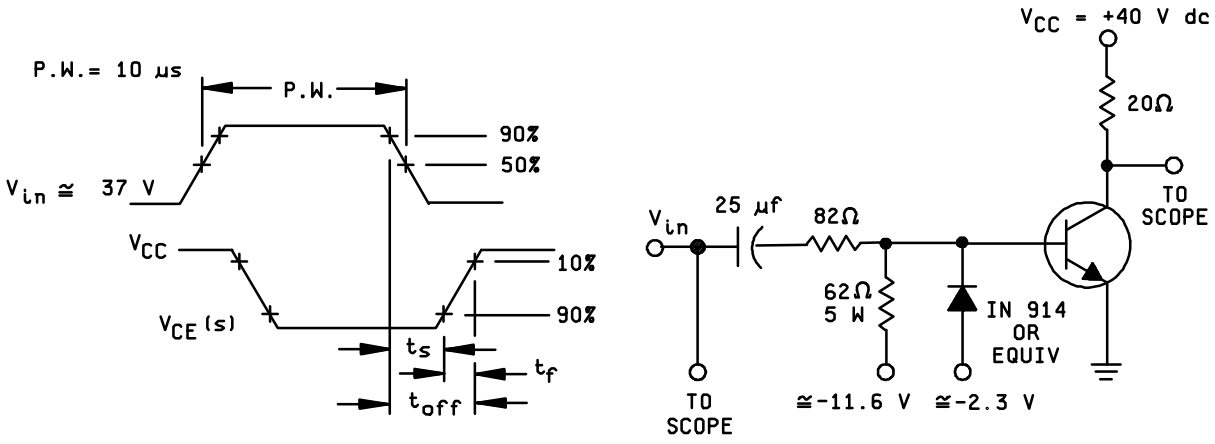
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NOTES:

1. The rise time ($t_r$) of the applied pulse shall be ≤ 20 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be $50^{\Omega}$.
2. Sampling oscilloscope: $Z_{in} \geq 1\ M^{\Omega}$, $C_{in} \leq 20$ pF, rise time ≤ 20 ns.
3. $t_{on}$ conditions: $I_C = 2$ A, $I_{B1} = 200$ mA.

FIGURE 4. Saturated turn-on switching waveform and time test circuit.




NOTES:

1. The rise time ($t_r$) of the applied pulse shall be ≤ 20 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be $50^{\Omega}$.
2. Sampling oscilloscope: $Z_{in} \geq 1\ M^{\Omega}$, $C_{in} \leq 20$ pF, rise time ≤ 20 ns.
3. $t_{on}$ conditions: $C_I = 2$ A, $I_{B1} = I_{B2} = 200$ mA.

FIGURE 5. Saturated turn-off switching time waveform and test circuit.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation (see 2.2.1).

b. The lead finish as specified (see 3.4.1).

c. Type designation and quality assurance level.

c. Packaging requirements (see 5.1).

e. For die acquisition, the JANHC or JANKC letter version shall be specified (see figures 2 and 3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer's List QML No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Application guidance. The following PNP type transistor is complimentary to the NPN device listed herein.

| NPN | PNP |
|---|---|
| 2N5339 | 2N6193 |

6.5 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example, JANHCA2N6193) will be identified on the QML.

| JANC ordering information | | |
|---|---|---|
| PIN | Manufacturers | |
| | 33178 | 34156 |
| 2N5339 | JANHCA2N5339<br>JANKCA2N5339 | JANHCB2N5339<br>JANKCB2N5339 |

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-A156)



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**STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL**

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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| **I RECOMMEND A CHANGE:** | **1. DOCUMENT NUMBER**<br>MIL-PRF-19500/560C | **2. DOCUMENT DATE (YYMMDD)**<br>990728 |
|---|---|---|

**3. DOCUMENT TITLE** SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING TYPE 2N5339 JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

**4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)**

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>Commercial<br>DSN<br>FAX<br>EMAIL | 7. DATE SUBMITTED<br>(YYMMDD) |

**8. PREPARING ACTIVITY**

| a. Point of contact: Alan Barone | b. TELEPHONE<br>Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil |
|---|---|
| c. ADDRESS:<br>Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000 | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>Defense Standardization Program Office (DLSC-LM)<br>8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221<br>Telephone (703) 767-6888 DSN 427-68880 |



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